Work Sample Note

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The attached paper applies the Synthetic Difference-in-Differences (SDID) method to evaluate the economic impact of the 1994 CFA franc devaluation across 14 West and Central African countries. The central question is whether the policy improved GDP growth in affected countries relative to a credible counterfactual.

What this sample demonstrates:

- **Data preparation and cleaning.** I constructed a 23-country, 15-year panel from the Penn World Table (PWT 10.0), matching countries by ISO code, computing log-differenced GDP growth rates, and structuring the data into the matrix format required for panel estimation.
- **Quantitative analysis and method comparison.** I applied three estimators side by side (Difference-in-Differences, Synthetic Control and SDID) and diagnosed why each produces a different answer. Plain DiD finds essentially no effect (-0.18 pp), because several control countries collapsed in civil wars just before the devaluation. SC (+2.52 pp) and SDID (+2.90 pp) reweight the comparison to correct for this.
- **Translation for broad audiences.** The paper is written to be accessible beyond econometricians. Visualizations show group-level GDP trends before and after the devaluation, and the discussion explains why methodological choice is itself a policy-relevant decision, not merely a technical one.
- **Substantive domain knowledge.** The CFA franc case draws on familiarity with West African macroeconomic data and institutional context, developed during 14 years compiling national accounts at Niger's National Institute of Statistics (INS Niger).

Key finding:

Once the comparison accounts for the conflict-driven collapse in several non-CFA countries, the 1994 devaluation is associated with roughly 2.5 to 3 percentage points higher annual GDP growth in the CFA zone over 1994–2000. A naive DiD comparison misses this effect entirely. The estimates are imprecise, however: with 14 treated and only 9 control countries, none of the confidence intervals excludes zero. This is consistent with the broader literature on the devaluation, which documents short-run competitiveness gains but uneven effects across member states.

Technical details:

Tools: R (dplyr, ggplot2, synthdid, pwt10)
Data source: Penn World Table 10.0 (Feenstra, Inklaar and Timmer, 2015)
Method: Arkhangelsky et al. (2021), Synthetic Difference-in-Differences
Code: https://github.com/msadou02-hue/cfa-devaluation-sdid

Causal Inference with Synthetic Difference-in-Differences

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1. Introduction

The paper *Synthetic Difference in Differences* by Dmitry Arkhangelsky, Susan Athey, David Hirshberg, Guido Imbens and Stefan Wager develops a novel estimator for evaluating causal effects using panel data in a setting where some units are exposed to the policy in some time periods but not others. The Synthetic Difference in Differences (SDID) method merges the logic of two popular frameworks:

- **Difference-in-Differences (DiD):** assumes parallel trends for treated and control groups.
- **Synthetic Control (SC):** builds a counterfactual for a treated unit from a weighted combination of control units.

SDID improves on each method's limitations, offering robustness, flexibility and improved performance across diverse empirical scenarios.

2. Motivation and Context

Causal inference in panel data settings often involves comparing treated and untreated units across time. This requires methods that can adjust for both observed and unobserved confounding and account for different pre-treatment trajectories. Two of the most widely used approaches are Difference-in-Differences (DiD) and Synthetic Control (SC). Each method has strengths and weaknesses that make them suitable for particular empirical scenarios but problematic in others. The SDID framework emerges from the motivation to reconcile these approaches and create a more robust estimator.

2.1 Difference-in-Differences (DiD)

Difference-in-Differences is a classical method for evaluating policy impacts when randomized experiments are not feasible. It compares the before-after difference in outcomes between treated and untreated groups. The identifying assumption is that, absent treatment, the average outcome trends of treated and control units would have moved in parallel over time.

This parallel trends assumption is strong and often unverifiable. If the pre-treatment trends of the control group do not align well with those of the treated group, the DiD estimator may yield biased results. Moreover, DiD is typically implemented through a two-way fixed effects regression, which assumes additive separability of unit and time effects. In practice, this assumption may not hold, particularly in settings with complex time-varying confounding or when treatment is assigned non-randomly.

2.2 Synthetic Control (SC)

The Synthetic Control method addresses some of DiD's shortcomings by explicitly constructing a counterfactual for treated units. This counterfactual is a weighted average of untreated units, where the weights are chosen to match the pre-treatment trajectory of the treated unit(s) as closely as possible. SC is most commonly used when only a small number of units are treated, often just one.

While SC improves the credibility of counterfactual comparisons by closely matching trends, it introduces new challenges. The method does not easily accommodate statistical inference using conventional tools. Additionally, SC is sensitive to the selection of pre-treatment periods and can produce unstable estimates if pre-treatment fit is poor or if the donor pool lacks appropriate comparators.

2.3 SDID: Combining Strengths

The SDID estimator blends the philosophies of DiD and SC. It retains the regression-based framework of DiD, which enables straightforward implementation and inference, but augments it with the synthetic matching feature of SC.

Specifically, SDID introduces unit and time weights to balance pre-treatment trajectories across units and periods. The unit weights ensure that treated units are compared to similar control units, akin to SC. The time weights emphasize the comparison of time periods with similar dynamics, mitigating bias from unusual time shocks.

This combination allows SDID to:

- reduce reliance on the strong parallel trends assumption by reweighting to create parallelism in expectation;
- provide valid statistical inference through regression techniques;
- generalize to settings with staggered or group-level treatment;
- improve estimation precision by emphasizing informative comparisons.

In essence, SDID formalizes what researchers often do informally, adjusting data to satisfy assumptions before applying DiD. By automating this process through a principled weighting scheme, SDID achieves greater transparency and robustness.

3. Methodology

3.1 Setup and Notation

Let $i = 1, \ldots, N$ index units and $t = 1, \ldots, T$ index time periods. Y_{it} is the observed outcome and $W_{it} \in \{0, 1\}$ the treatment indicator. The first N_{co} units are controls and the remaining $N_{tr} = N - N_{co}$ are treated; treatment starts after period T_{pre}, leaving $T_{post} = T - T_{pre}$ post-treatment periods. μ is the grand mean, α_i the unit fixed effects and β_t the time fixed effects.

3.2 SDID Estimator

The SDID estimate is obtained by solving a weighted two-way fixed effects regression:

$$\left(\hat{\tau}^{sdid}, \hat{\mu}, \hat{\alpha}, \hat{\beta}\right) = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T} \left(Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau\right)^2 \hat{\omega}_i \hat{\lambda}_t$$

DiD is the special case with equal weights for all units and periods; SC corresponds to dropping the unit fixed effects and the time weights.

3.3 Weighting Strategy

Unit weights $\hat{\omega}$ are chosen so that the weighted control units track the average treated unit before treatment:

$$\ell_{unit}(\omega_0, \omega) = \sum_{t=1}^{T_{pre}} \left(\omega_0 + \sum_{i=1}^{N_{co}} \omega_i Y_{it} - \frac{1}{N_{tr}} \sum_{i=N_{co}+1}^{N} Y_{it}\right)^2 + \zeta^2 T_{pre} \|\omega\|_2^2$$

Time weights $\hat{\lambda}$ balance pre- and post-treatment periods for the control units:

$$\ell_{time}(\lambda_0, \lambda) = \sum_{i=1}^{N_{co}} \left(\lambda_0 + \sum_{t=1}^{T_{pre}} \lambda_t Y_{it} - \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^{T} Y_{it}\right)^2$$

Both sets of weights are non-negative and sum to one. The intercepts ω_0 and λ_0 allow for level differences, which the fixed effects then absorb.

4. Case Study: CFA Franc Devaluation

In this section, we apply SDID to the 1994 devaluation of the CFA franc. This shock affected 14 African countries and provides a natural policy intervention to test causal inference methods. We examine real GDP growth, comparing SDID with DiD and SC.

Context. The 1994 CFA franc devaluation was a major economic reform in which the value of the CFA franc was halved relative to the French franc. It affected all countries using the CFA franc, in West and Central Africa, with the goal of improving external competitiveness by boosting exports and reducing trade imbalances.

This natural experiment lends itself well to SDID because:

- it has a clear treatment timing (January 1994);
- the treated units are well defined: the countries of the CFA franc zone;
- suitable control countries exist in West and Central Africa that were not part of the CFA zone;
- longitudinal macroeconomic data (1985–2000) are available for both treated and control countries.

We construct a panel of real GDP growth for these countries and estimate the effect of the devaluation using DiD, SC and SDID.

Data Preparation

Countries are matched by ISO code rather than by name, because the Penn World Table spells several names differently (for example "Congo" for the Republic of Congo). Matching on names silently drops those countries.

```
library(dplyr); library(ggplot2); library(pwt10); library(synthdid)

# CFA zone: WAEMU (West Africa) and CEMAC (Central Africa)
cfa_iso <- c("BEN", "BFA", "CIV", "GNB", "MLI", "NER", "SEN", "TGO",
             "CMR", "CAF", "TCD", "COG", "GNQ", "GAB")
# Neighbouring non-CFA economies
control_iso <- c("GHA", "NGA", "LBR", "SLE", "GIN", "COD",
                 "AGO", "GMB", "MRT")

data("pwt10.0")
df <- pwt10.0 %>%
  filter(isocode %in% c(cfa_iso, control_iso), year >= 1985, year <= 2000) %>%
  transmute(iso = as.character(isocode), country = as.character(country),
            year, rgdpna) %>%
  arrange(iso, year) %>%
  group_by(iso) %>%
  mutate(gdp_growth = 100 * (log(rgdpna) - log(lag(rgdpna)))) %>%
  ungroup() %>%
  filter(!is.na(gdp_growth)) %>%
  mutate(cfa = iso %in% cfa_iso,
         treated = as.integer(cfa & year >= 1994))

# Keep a balanced panel
panel <- df %>% group_by(iso) %>% filter(n() == n_distinct(df$year)) %>% ungroup()
```

`synthdid` requires the outcome matrix to list control units first and treated units last. `panel.matrices()` builds the matrix in that order and labels its rows and columns:

```
panel_df <- as.data.frame(select(panel, country, year, gdp_growth, treated))
setup <- panel.matrices(panel_df, unit = "country", time = "year",
                        outcome = "gdp_growth", treatment = "treated")
```

```
dim(setup$Y)   # countries x years
```

```
## [1] 23 15
```

```
c(N0 = setup$N0, T0 = setup$T0)
```

```
## N0 T0
##  9  8
```

The panel contains 14 CFA countries and 9 controls observed over 15 years (1986–2000), with 8 pre-treatment years.

Visualize the Panel Structure

```
group_avg <- panel %>%
  group_by(year, group = ifelse(cfa, "CFA zone", "Non-CFA")) %>%
  summarise(avg_growth = mean(gdp_growth), .groups = "drop")

ggplot() +
  geom_line(data = panel, aes(year, gdp_growth, group = country),
            colour = "grey75", linewidth = 0.4) +
  geom_line(data = group_avg, aes(year, avg_growth, colour = group), linewidth = 1.2) +
  geom_vline(xintercept = 1993.5, linetype = "dashed") +
  scale_colour_manual(values = c("CFA zone" = "#1f5aa6", "Non-CFA" = "#c0392b")) +
  coord_cartesian(ylim = c(-20, 25)) +
  labs(title = "Real GDP growth, 1986-2000",
       subtitle = "Dashed line marks the January 1994 CFA franc devaluation",
       x = NULL, y = "Real GDP growth (%)", colour = NULL) +
  theme_minimal(base_size = 12) + theme(legend.position = "bottom")
```

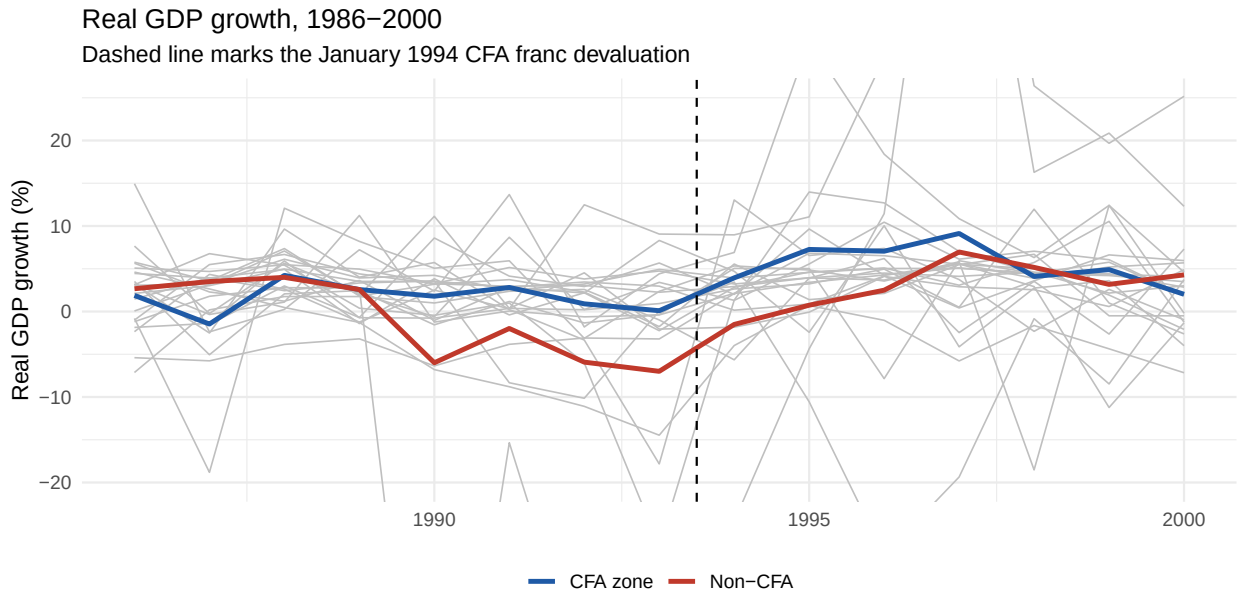


Figure 1: Real GDP growth by country (grey) and group averages.

The figure already shows the main identification problem. Before 1994 the two groups do not move in parallel: average non-CFA growth falls sharply in 1990–93, driven by civil wars in Liberia and Sierra Leone and conflict in Angola and the Democratic Republic of the Congo, and then recovers in the mid-1990s, at the same time as the devaluation.

Estimate Effects

```
did_est  <- did_estimate(setup$Y, setup$N0, setup$T0)
sc_est   <- sc_estimate(setup$Y, setup$N0, setup$T0)
sdid_est <- synthdid_estimate(setup$Y, setup$N0, setup$T0)

# Jackknife standard errors (leave one country out). Placebo standard errors
# are not available because there are more treated than control countries.
se <- sapply(list(did_est, sc_est, sdid_est),
             function(x) sqrt(vcov(x, method = "jackknife")))
```

Table 1: Estimated effect of the devaluation on real GDP growth (percentage points), with jackknife standard errors and 95% confidence intervals.

Estimator	Estimate	SE	CI low	CI high
DiD	-0.18	4.93	-9.85	9.49
SC	2.52	1.91	-1.22	6.25
SDID	2.90	2.85	-2.69	8.48

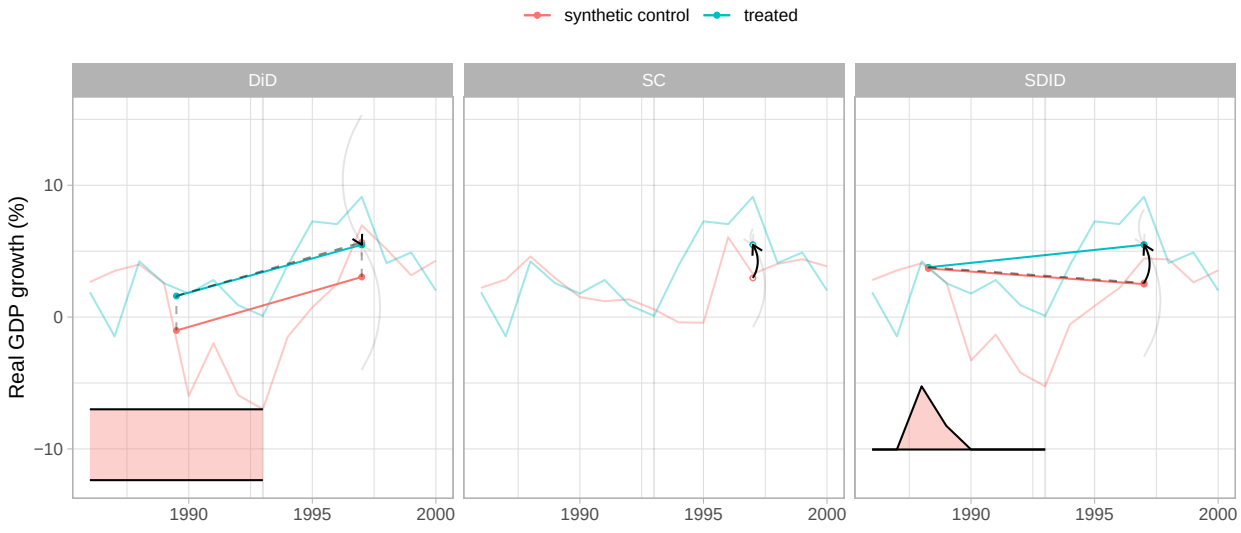


Figure 2: Treated and synthetic control trajectories for each estimator. Shaded areas at the bottom show the time weights; arrows show the estimated effect.

Why the Estimators Disagree

- **DiD (-0.18 pp).** DiD weights all countries and all pre-treatment years equally. Because non-CFA growth was depressed by conflict in 1990–93 and then recovered, the control group's improvement after 1994 is as large as the CFA zone's, and DiD finds essentially no effect.
- **SC (+2.52 pp).** SC reweights the control countries so that their combined growth path tracks the CFA zone before 1994, which removes much of the distortion.
- **SDID (+2.90 pp).** SDID also reweights the pre-treatment *years*. It places 100% of the time weight on 1988 and 1989, before the conflicts, so the baseline is a period when both groups were growing normally. The unit weights are close to equal (between 0.08 and 0.12), so no single control country drives the result.

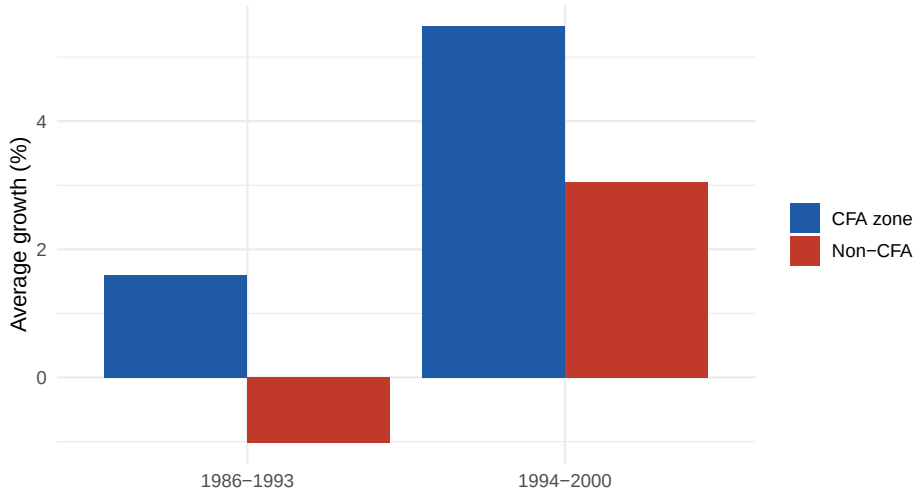


Figure 3: Average real GDP growth before and after the devaluation.

The simple averages illustrate the same point. CFA growth rose from 1.60% to 5.48% a year, but non-CFA growth rose by a similar amount, from -1.02% to 3.04%, largely as a recovery from conflict rather than as a counterfactual for the CFA zone.

5. Example from the Paper: California Proposition 99

To illustrate SDID on a well-known benchmark, we replicate the case used in Arkhangelsky et al. (2021): California's Proposition 99, a 1989 anti-smoking law that significantly raised cigarette taxes and imposed new restrictions. We compare California's cigarette consumption before and after 1989 with a counterfactual built from other U.S. states that did not adopt similar policies.

```
data("california_prop99")
setup99 <- panel.matrices(california_prop99)

est99 <- list(DiD  = did_estimate(setup99$Y, setup99$N0, setup99$T0),
              SC   = sc_estimate(setup99$Y, setup99$N0, setup99$T0),
```

```
              SDID = synthdid_estimate(setup99$Y, setup99$N0, setup99$T0))
round(sapply(est99, as.numeric), 2)
```

```
##    DiD     SC   SDID
## -27.35 -19.62 -15.60
```

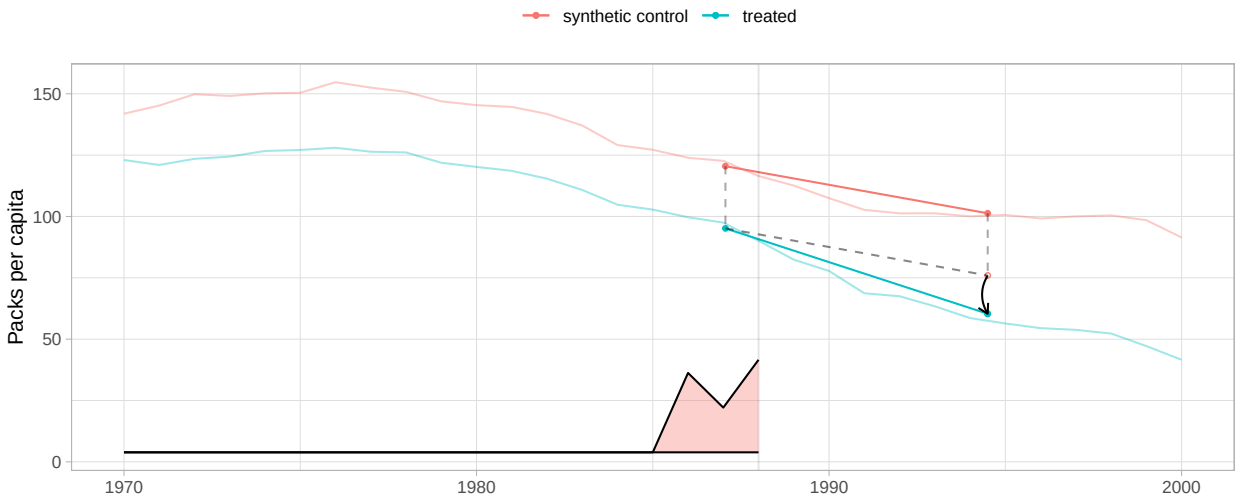


Figure 4: Proposition 99: California versus its synthetic counterfactual.

The estimates reproduce Table 1 of the original paper. SDID gives a smaller and more credible effect than DiD because it matches California's pre-1989 trajectory before comparing changes.

6. Discussion

The SDID approach provides advantages in cases where traditional DiD fails to balance pre-treatment trends. The CFA application is a clear example. The control countries' growth path before 1994 was distorted by civil wars, so the parallel trends assumption fails and plain DiD finds no effect at all. SC and SDID, which reweight countries (and, for SDID, years) to build a comparable baseline, both point to a positive effect of about 2.5 to 3 percentage points of annual GDP growth.

The choice of method is therefore not a technical detail: it determines whether the evaluation concludes that the devaluation did nothing or that it substantially supported growth. The uncertainty is large, however. With 14 treated countries and only 9 controls, the jackknife standard errors are between 1.91 and 4.93 percentage points and none of the estimates is statistically significant at the 5% level.

7. Conclusion

SDID is a powerful causal inference tool for policy evaluation with panel data. By combining reweighted outcomes with fixed effects, it supports more credible identification under weaker

assumptions than DiD. Applied to the CFA franc devaluation, it shows that accounting for pre-treatment differences changes the conclusion: the devaluation is associated with higher GDP growth in the CFA zone, an effect that a naive DiD comparison misses. A larger donor pool, or outcomes measured at a finer level, would be needed to estimate the effect precisely.

References

- Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., & Wager, S. (2021). Synthetic Difference-in-Differences. *American Economic Review*, 111(12), 4088–4118.
- Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic Control Methods for Comparative Case Studies. *Journal of the American Statistical Association*, 105(490), 493–505.
- Bertrand, M., Duflo, E., & Mullainathan, S. (2004). How Much Should We Trust Differences-in-Differences Estimates? *Quarterly Journal of Economics*, 119(1), 249–275.
- Feenstra, R. C., Inklaar, R., & Timmer, M. P. (2015). The Next Generation of the Penn World Table. *American Economic Review*, 105(10), 3150–3182.